Personalized Federated Contrastive Learning for Recommendation

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***Abstract*—Recommender systems play crucial roles in addressing the issue of information overload, but traditional centralized storage in recommendation poses significant privacy concerns. In recent years, federated learning has been successfully introduced into a recommendation, while these algorithms still encounter several challenges. First, real-world recommendation scenarios often suffer from sparse data, making it difficult for models to learn reliable representations. Second, data heterogeneity necessitates the design of personalized models to enhance recommendation performance. To address these challenges, we propose a federated recommendation approach based on graph neural networks, named federated personalized contrastive learning for recommendation. On the client side, we propose a contrastive learning approach to enhance the embedding quality of nodes (users or items) by maximizing positive similarities. Specifically, we formulate the concept of structural neighbors based on the graph structure and devise a contrastive learning objective. We treat nodes and their structural neighbors as positive pairs to better learn node representations. On the server side, we group users based on the learned representations and compute cluster-level federated models and a global model. Each user learns a personalized model by combining these two models. Extensive experiments on five real-world datasets demonstrate that the proposed algorithm outperforms existing methods in terms of performance.**

***Index Terms*—Contrastive learning, federated learning, graph neural network (GNN), graph structure, personalized recommendation.**

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I. Introduction

With the rapid development of the Internet, the amount of information present in our lives has far exceeded the needs and processing capacity of individuals. This phenomenon is commonly known as information overload. Recommendation systems have emerged as an effective solution to alleviate information overload by helping users to navigate through the overwhelming amount of information (e.g., music, movies, and news) and find what they specifically need [1], [2], [3].

Early recommendation systems mainly adopted collaborative filtering, which works by recommending items to users based on their past preferences or clicks [4]. If a user has similar tastes or preferences to another user in the past, it is likely that they will have similar interests in the future [5]. However, traditional methods struggle to capture the nonlinear relationships in recommendation systems. In recent years, researchers have applied graph neural networks (GNNs) to recommendation systems [6], [7], [8]. GNN models the interaction data as a graph and captures high-order information between users and items, thereby further improving the performance of recommendation systems. Centralized recommendation approaches can lead to serious practical challenges [9], e.g., communication costs and risks of privacy violation of user data. Additionally, due to the strict privacy regulations, such as general data protection regulation (GDPR), centralized data storage may not be the preferred choice for future recommendation systems.

Federated learning is a privacy-preserving distributed learning approach proposed by Google [10], where the participants' actual data can be stored locally on each client device, and the model can be updated by sharing intermediate parameters such as model parameters and gradients with a central server [11]. In recent years, there has been a significant amount of research into federated recommendation algorithms. Many classic recommendation algorithms have been extended to federated settings, such as federated collaborative filtering [12], [13], federated matrix factorization [14], [15], and federated GNNs [16], [17]. Perifanis and Efraimidis [13] presented a framework called federated neural collaborative filtering (FedNCF). FedNCF allows clients to update the network weights and user/item profiles locally. Hu et al. [17] proposed a GNN-based federated recommendation framework called P-GCN, which can utilize decentralized user–item interaction graphs to construct GNN models in a privacy-aware manner. However, existing federated recommendation methods still face

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two major problems. The first is personalization. Traditional federated averaging (FedAvg) methods are designed for independent identically distribution (IID) data. In practical scenarios, however, data are often non-IID due to varying user preferences. Personalization refers to providing specific recommendation models to users rather than using the same aggregated model for all clients [18], [19]. This can lead to suboptimal model performance. Personalized recommendations can mitigate the reduction in recommendation performance caused by non-IID or heterogeneous nature of data [20]. Zhang et al. [21] proposed a personalized federated framework and designed a dual personalization mechanism to learn fine-grained personalization for users and items. ReFRS [22] is a resource-efficient federated recommendation approach, which achieves decentralized recommendations with dynamic and diverse user preferences. Although research on personalized recommendations has made progress, it does not simultaneously address model personalization and data sparsity. The second challenge is that of data sparsity. In federated recommendation systems, user data are typically scattered and stored separately, resulting in smaller datasets for individual users that are insufficient for comprehensive training. Furthermore, user data are often non-IID, leading to more sparse and potentially noisy training data for the model. This challenge makes it difficult for federated models to learn reliable representations, thereby affecting recommendation performance and user experience, particularly for GNNs which are more susceptible to the effects of sparsity [23]. Contrastive learning has been proven to be an effective solution to address the issue of data sparsity [24]. Many central recommendation algorithms and federated recommendation algorithms mitigate the impact of data sparsity by designing specific contrastive learning objectives [25], [26]. Federated contrastive learning (FedCL) [25] proposed a hard negative sampling method that utilizes high-quality negative samples to design contrast pairs for effective model training. Dual-contrastive for federated social recommendation (DFSR) [26] proposed a federated recommendation framework that minimizes the distance between users and their trusted users in feature space and maximizes the distance between local and global items. However, they do not consider the design of contrast pairs from the structure of GNNs in federated recommendation.

Inspired by the aforementioned issues, we propose a personalized model in a federated setting and exploit the relationships between nodes of GNNs in federated recommendation clients, so as to design an effective contrastive learning objective to improve node representation learning. Therefore, we propose a new federated learning framework based on GNNs, called federated personalized contrastive learning framework (FedPCL). First, we define structural neighbors as nodes connected through high-order paths in the GNNs. On the client side, we leverage contrastive learning methods to minimize the distance between a node's representation and its structural neighbors, which effectively captures high-order information from the GNN and improves node representations, mitigating the effects of data sparsity in federated recommendation clients. Next, FedPCL groups users into different clusters based on their learned representations. Then, a cluster-level model is then built for each cluster. Clients combine the global and cluster-level models to obtain personalized models for recommendations. Experiments validate the effectiveness of the algorithm.

Our main contributions can be summarized as follows.

1) Considering federated personalized recommendation and data sparsity, we propose a novel federated personalized contrastive learning framework, called FedPCL. This framework incorporates structural neighbors to learn better node representations and designs a personalized recommendation model.
2) We design structural contrastive learning objects based on the structure of GNNs, allowing participants to incorporate high-order structural neighbors to improve recommendation performance and mitigate the effects of data sparsity in federated learning clients.
3) To address the challenges posed by data heterogeneity and non-IID data, we propose to combine cluster-level models and a global model to provide personalized recommendations to users.
4) We conduct extensive experiments on five public datasets and show that our approach consistently outperforms other baseline methods.

The remainder of this article is structured as follows. Section II provides a brief overview of related work. Section III provides a detailed description of the proposed FedPCL. Section IV introduces and analyzes the performance of the proposed algorithm on five public datasets. Finally, in Section V, we conclude the article.

II. Related Work

A. GNN for Recommendation

In recent years, GNN techniques have been widely applied in diverse fields, for example, emotion recognition [27] and recommendation [28]. Due to the fact that a significant portion of information in recommender systems inherently exhibits a graph structure, and GNN holds an advantage in learning graph representations [1], [29]. To achieve diversification in GNN-based recommender systems, Yang et al. [30] proposed a method by directly improving the embedding generation process, called DGRec. It integrates submodular neighbor selection, layer attention, and loss reweighting into the GNN for diversified recommendation. Liu et al. [23] proposed a self-supervised GNN for sequential recommendation, encoding the short-term graphs based on time intervals, and utilized GNN to learn short-term collaborative relationships, thereby enhancing the accuracy and robustness of sequential recommendation. He et al. [8] simplified the GCN algorithm to make it more concise and suitable for recommendation scenarios, named LightGCN. It learns embeddings for users and items by performing linear propagation on the user–item interaction graph and uses the weighted sum of embeddings learned across all layers as the final embedding. Wang et al. [7] proposed a new social recommendation model based on context-aware GNNs, called CENTRIC, which incorporates user context collaboration and tensor decomposition to further enhance user feature representations. Liu et al. [6] proposed a novel method, called SAGLG, by considering collaborative embeddings and group embeddings simultaneously. SAGLG designs a latent group mining

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module to generate group embeddings for users and items. The collaborative embedding module employs graph embedding techniques to provide semantic collaborative embeddings for users and items. The self-attention graph convolution module learns fine-grained representations of nodes. Our algorithm also leverages the GNN model to learn user/item embeddings from user–project interactions.

B. Contrastive Learning

Contrastive learning, as a kind of self-supervised learning methods, has been widely applied in various fields such as computer vision (CV) [31], natural language processing (NLP) [32], and graph learning [33]. The basic idea of contrastive learning is to select appropriate positive and negative pairs, maximizing the agreement between positive pairs and minimizing the agreement between negative pairs [34]. Zhang et al. proposed an adaptive-margin-enhanced contrastive learning technique to learn the global model, thereby enhancing interclass separability [35]. In the field of recommender systems, Hao et al. treated the structural and semantic information of user–item graphs as two contrastive views for recommendation and then proposed a contrastive learning model for graph collaborative filtering [36]. It generates adaptive contrastive views through a metaoptimized two-step training strategy and enhances the recommendation performance of the model. Yu et al. created contrastive views by adding uniform noise to the embedding space [24]. To reduce computational complexity, this method replaces the final layer contrast with cross-layer contrast. Lin et al. [37] presented a prototype-contrastive objective to capture contextual information. Specifically, for each item or user, the positive sample is the prototype of the cluster to which it belongs, while the negative sample is the prototype of other clusters. The prototype represents the cluster center and the prototype-contrastive objective is learned using the expectation–maximization (EM) algorithm. Additionally, this algorithm applies cross-layer contrasting, treating the corresponding representations output from the even-numbered layer GNN as positive samples for each user or item.

C. Federated Recommendation

In recent years, there has been extensive research on the application of federated learning in privacy-preserving recommender systems [38]. Federated collaborative filtering (FCF) [12] is the first method that extends collaborative filtering to the federated setting. It updates local models using stochastic gradient descent and updates the global model using FedAvg [10]. To preserve local data privacy, Chai et al. [14] adopted a distributed matrix factorization algorithm and designed homomorphic encryption technology to upload encrypted gradients to the server. Wu et al. [38] utilized user–item graph expansion techniques to identify neighbors with common interaction items, enabling each user to maintain an extended local subgraph. To address the fairness and personalization issues in federated recommendation, we proposed fair and personalized federated recommendation (FPFR) [39]. It designs a soft attention mechanism at the client side and integrates a filtering network to better learn user preferences. Additionally, it introduces personalized models and a fairness loss function for model training. Recently, there has been research on applying contrastive learning to federated recommendation. Wu et al. proposed a hard negative sampling method that utilizes high-quality negative samples for effective model training [25]. Luo and Liu proposed a federated social recommendation framework based on contrastive learning. This method minimizes the distance between users and their trusted users in the user-level feature space and maximizes the consistency between local and global item embeddings [26]. To address the statistical heterogeneity in federated recommender systems, He et al. [40] proposed a novel collaborative clustering federated recommendation mechanism called CoFedRec. It groups users based on specific item categories for learning and incorporates a contrastive learning loss on the client side, which is based on item membership generated by the server side. However, existing methods construct contrastive pairs through specific sampling strategies without considering the potential structural relationships in the GNN architecture.

III. Methodology

A. Problem Formulation

Before describing the proposed method, we first introduce some important symbols used in this study and formalize our main task.

We define a target recommender system to contain a set of users $\mathcal{U} = \{u_1, u_2, \ldots, u_N\}$, and a set of items $\mathcal{V} = \{v_1, v_2, \ldots, v_M\}$, where N and M are the numbers of users and items, respectively. The user–item interaction matrix $\mathbb{R} = [r_{uv}]_{M\times N}$ is used to represent user's ratings for items. We consider the common implicit feedback scenario, if user u has clicked on item v, then $r_{uv} = 1$ represents that user u likes the item v, otherwise $r_{uv} = 0$. We construct a bipartite user–item graph $\mathcal{G}$ based on the interaction matrix $\mathbb{R}$, where each user is connected to all the items they have interacted with. Suppose there is a user u_i who has clicked on k items in total, we can represent all the items that have been evaluated using $\mathcal{V}_i = \{v_1, v_2, \ldots, v_k\}$. In our federated learning scenario, there is one server and N clients, where each client maintains a local subgraph $\mathcal{G}_i$. To protect user privacy, the data of each user cannot leave the local client. Therefore, the server cannot directly access the user's interaction data but only the model parameters. Our goal is to utilize GNN to obtain user embedding matrix $e_u \in \mathbb{R}^d$ and item embedding matrix $e_i \in \mathbb{R}^d$ in a privacy-preserving manner, where d represents the dimension of the embeddings. Then, we calculate the predicted preference scores based on the formula $r_{ui} = e_u e_i$ and recommend a list of top-K items to the user.

B. GNN in Recommendation

In many recommendation scenarios, data can be represented as graph structures. GNNs have advantages in graph representation learning, which models the interaction data as graphs and then captures high-order interactions between nodes on the graph through multilayer graph convolution operations. Therefore, the GNN utilized in the proposed FedPCL framework is described in detail as follows.

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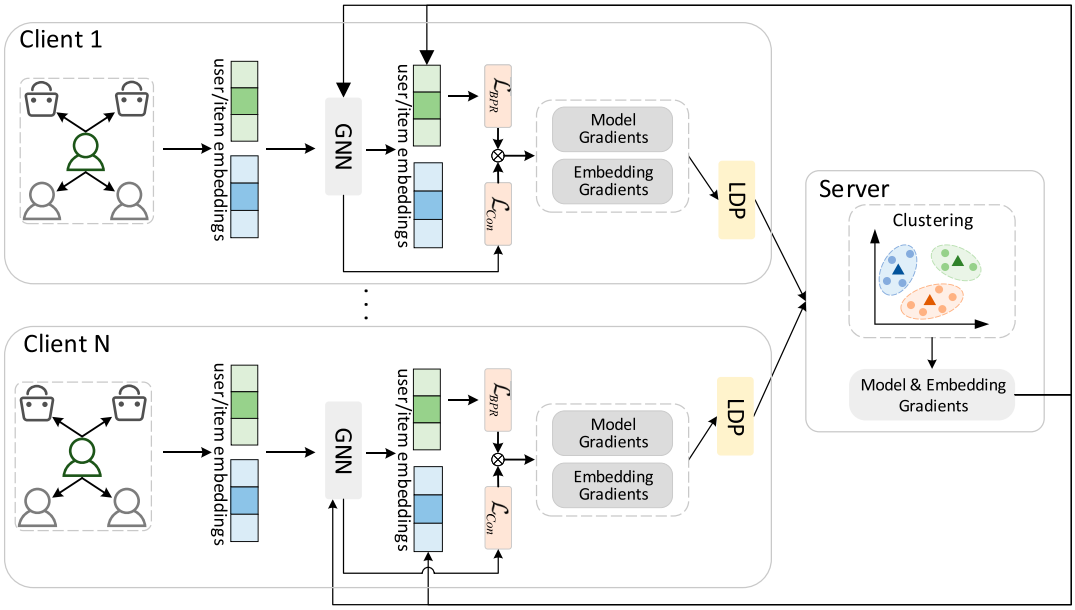

Fig. 1. Framework of our proposed FedPCL.

1) Initialization: Initialize the user embeddings and item embeddings as e_u and e_i, respectively. These embeddings represent the features of the nodes and are often randomly initialized or using pretrained embeddings.

2) Propagation: This stage aggregates the neighbor information of the kth layer to obtain the node representation of the $(k+1)$th layer

$$\begin{aligned} e_u^{(k+1)} &= f_{\text{propagate}}\left(\left\{e_i^{(k)} | i \in \mathcal{N}_u\right\}\right) \\ e_i^{(k+1)} &= f_{\text{propagate}}\left(\left\{e_u^{(k)} | u \in \mathcal{N}_i\right\}\right) \end{aligned} \tag{1}$$

where $\mathcal{N}_u$ represents the set of neighbors of user u in the interaction graph $\mathcal{G}_u$, and k represents the layer of the GNN. For user u, the propagation function aggregates the representations of its kth layer neighbors to generate the node representation of the $(k+1)$th layer. After multiple iterations of aggregation, the node representation of the kth layer aggregates information from k-hop neighbors.

3) Update: After K layers of aggregation, we apply the update function to combine the embeddings $\left[e_u^{(0)}, e_u^{(1)}, \ldots, e_u^{(K)}\right]$ and $\left[e_i^{(0)}, e_i^{(1)}, \ldots, e_i^{(K)}\right]$ obtained at each layer and form the final representation of the nodes

$$\begin{aligned} e_u &= f_{\text{update}}\left(\left[e_u^{(0)}, e_u^{(1)}, \ldots, e_u^{(K)}\right]\right) \\ e_i &= f_{\text{update}}\left(\left[e_i^{(0)}, e_i^{(1)}, \ldots, e_i^{(K)}\right]\right). \end{aligned} \tag{2}$$

4) Prediction: Utilize updated nodes e_u and e_i for prediction. Calculate the likelihood of interaction between user u and item i

$$\hat{r}_{ui} = f\left(e_u, e_i\right) \tag{3}$$

where $f(\cdot)$ is a rating prediction function, the obtained value represents the probability of interaction between item i and user u.

5) Optimization: The BPR loss function [41] has been widely used in recommendation scenarios with implicit feedback. In this loss function, the goal is to learn the ranking of items for each user by maximizing the likelihood of observing the user's preference for one item over another. Let us consider a user u, BPR loss function can be represented as follows:

$$\mathcal{L}_{\text{BPR}} = -\sum_{(u,i,j)\in\mathcal{D}} \ln \sigma\left(\hat{r}_{ui} - \hat{r}_{uj}\right) \tag{4}$$

where $\mathcal{D}$ refers to the training dataset, i represents interacted items, j represents unrated items, and $\sigma(\cdot)$ is the sigmoid function.

C. FedPCL Framework Overview

As shown in Fig. 1, we propose a personalized federated GNN framework to address the challenges of non-IID data and data sparsity in recommendation scenarios while also reducing communication costs. Specifically, our framework consists of a trusted central server and N distributed clients. The central server is responsible for coordinating the entire model training process and aggregating the gradients received. Each client maintains a local subnetwork $\mathcal{G}_i$. We extend the local subnetwork using a similar approach to federated graph neural network (FedGNN) [38], which includes the user–item interaction data stored locally and the neighbors of each user.

The training process of FedPCL is as follows.

1) The server randomly selects a certain number of clients N for training and sends the item embeddings and model parameters to them.
2) The selected clients learn user/item embeddings and GNN model parameters based on their local subgraphs. Subsequently, calculate structural contrastive loss based on the output of l layers of the local network and calculate BPR loss based on the interaction dataset $\mathcal{D}$. Then, these

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two losses are combined to calculate the gradients for updating embeddings and the model.

3) Apply LDP to the updated gradients and send them to the server.
4) The central server clusters users into K groups in preparation for subsequent aggregation operations.
5) The server updates both the global model and cluster-level models separately and updates the item embeddings.
6) The server sends back the latest parameters to the clients for the next round of training.

This process repeats iteratively until the model converges. Next, we describe the details of each core step.

D. The Client-Side GNN

1) Structural Neighbor Contrastive Learning: Existing federated recommendation methods mainly rely on the observed interaction data for recommendation. However, there is still a lot of information on the higher order relationships between users and items on the GNN [37]. For example, users are more likely to purchase items similar to those preferred by their neighbors. Additionally, recommendation systems often suffer from data sparsity [42], leading to suboptimal performance. We aim to tackle this issue by fully exploiting the latent relationships between users and items, such as user similarities and item correlations. The mechanism of contrastive learning is to pull closer the distance between semantically similar nodes (positive sample pairs) in the embedding space while pushing apart the distance between semantically dissimilar nodes (negative sample pairs). This approach can effectively learn the similarity between nodes by leveraging the structural information, even in the presence of sparse data, thus enabling accurate capture of the underlying information. Federated recommendations often suffer from a decline in model performance due to privacy protection constraints. By applying contrastive learning to the federated recommendation scenario, we can significantly enhance the performance of federated recommendations, making it closely comparable to centralized training. Furthermore, compared to centralized contrastive learning, our approach reduces storage requirements and eases the computational burden on the server. This not only preserves user privacy but also achieves effective recommendation performance. Therefore, we propose a structure-contrastive method based on graph networks to extract potential information by mining the relationships between nodes (i.e., users/items).

We define the structural neighbors as the set of nodes $\mathcal{D}$ connected to node a via higher order paths. Benefiting from the characteristics of neural networks, the output of the kth layer of the GNN contains aggregated information from k-hop neighbors. Thus, we consider the output of the kth layer of the GNN as the representation of the k-hop neighbors of a node. Fig. 2 illustrates our proposed structural contrastive learning method. Because the interaction graph $\mathcal{G}$ is a bipartite graph, a GNN-based model can naturally aggregate information from homogeneous structural neighbors during even iterations. We can obtain representations of the homogeneous neighbors of users or items based on the outputs from even-numbered layers.

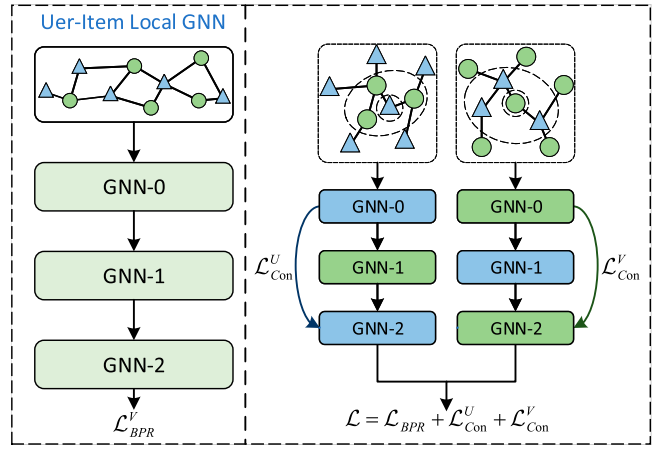

Fig. 2. Structure-contrastive learning. A comparison between traditional GNN and our proposed structural contrastive learning method. The circles represent user nodes, and triangles represent item nodes. The left subgraph illustrates the traditional three-layer GNN structure. The right subgraph represents our proposed structural contrastive learning approach.

These representations allow us to effectively model the relationship between users/items and their homogeneous structural neighbors.

For user u, we consider the homogeneous nodes on even layers of the local GNN as positive samples, and the homogeneous neighboring nodes of other users to be negative samples. Using the InfoNCE [34], we propose a structure-contrastive learning objective to minimize the distance between these embeddings

$$\mathcal{L}^{U}_{\text{Con}} = -\sum_{i\in\mathcal{U}} \log \frac{\exp\left(e_i^{(l)} \cdot e_i^{(0)}/\tau\right)}{\sum_{j\in\mathcal{U}} \exp\left(e_i^{(l)} \cdot e_j^{(0)}/\tau\right)} \tag{5}$$

where $e_i^{(0)}$ represents the initial features or embeddings of the user i, $e_i^{(l)}$ represents the normalized output of user i at the lth layer of the GNN, l is an even number. $e_i^{(l)} \cdot e_i^{(0)}$ represents the dot product similarity between them and τ is the temperature parameter.

We obtain the structural contrastive loss on the item side in a similar manner

$$\mathcal{L}^{V}_{\text{Con}} = -\sum_{v\in\mathcal{V}} \log \frac{\exp\left(e_v^{(l)} \cdot e_v^{(0)}/\tau\right)}{\sum_{n\in\mathcal{V}} \exp\left(e_v^{(l)} \cdot e_n^{(0)}/\tau\right)}. \tag{6}$$

We incorporate the structural contrastive learning loss with the recommendation loss in (4) to obtain the final loss function. The final loss function is represented as

$$\mathcal{L} = \mathcal{L}_{\text{BPR}} + \beta_1(\mathcal{L}^{U}_{\text{Con}} + \lambda\mathcal{L}^{V}_{\text{Con}}) + \beta_2||\Theta||_2 \tag{7}$$

where β_1 and β_2 are hyperparameters that control the weight of the structural contrastive loss and the regularization term, respectively. θ represents GNN model parameters.

2) Personalized Prediction: Traditional federated learning recommendation algorithms calculate a global model by simply averaging multiple local models. It is not until the final stages of training that the recommendation model achieves satisfactory performance. In this process, clients need to engage in multiple

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rounds of communication and consume a significant amount of computational resources. Moreover, recommendation methods based on FedAvg do not take non-IID data into consideration, which increases the discrepancy between local and global models and reduces model performance [43]. To reduce communication costs and mitigate the impact of non-IID data, we propose a multicentered personalized aggregation strategy that enables users to train personalized recommendation system models more quickly in a federated setting.

Specifically, we design a multicenter personalized recommendation model. This model consists of two modules: cluster models and a global model. First, based on the assumption that data from similar users are independently and identically distributed, we believe that similar users will benefit from similar models [44]. Recommendation based on users' preferences within the same cluster can accelerate model convergence. Therefore, the server uses clustering algorithms (e.g., K-means) to group similar users together, and within each group, calculating the cluster model with FedAvg. However, if only cluster models are used, there is a lack of information from the overall user population, which may lead to model overfitting. Moreover, due to limited data on clients' devices, the parameters of the cluster models may not be the optimal. Therefore, we use the FedAvg algorithm to integrate all cluster models into a global model. Subsequently, the server sends these two sets of models to the clients, who then use weighted combination to obtain the local model.

At time step t, the global model received by user u from the server is represented as h^t_{global}, and the cluster model is denoted as $h^t_{C(u)}$, where $C(u)$ is the cluster containing user u. We leverage these two models to construct a personalized model

$$h^t_u = \mu_1 h^t_{C(u)} + \mu_2 h^t_{\text{global}} \tag{8}$$

where μ_1 and μ_2 are hyperparameters for weighting. The user u utilizes the personalized model h^t_u for recommendations.

3) Local Differential Privacy (LDP): Federated matrix factorization (FedMF) has theoretically shown that in the recommendation scenario of federated matrix factorization, knowing a user's gradients uploaded consecutively twice can lead to the inference of the user's rating information [14]. Despite the complexity of our GNN compared to matrix factorization, uploading gradients still poses privacy risks. To protect users' privacy, we use LDP module to encrypt the gradients. LDP is a commonly used privacy protection technique in federated recommendation systems [9], with the basic idea of adding noise to hide individual-specific information while ensuring the accuracy of data analysis results. Specifically, we represent the item embedding gradient of client u as g^u_i and the model gradient as g^u_m. For each gradient $g^u \in \{g^u_i, g^u_m\}$, we apply the LDP module as follows:

$$\tilde{g}^u = \text{clip}(g^u, \sigma) + \text{Laplacian}\,(0, \lambda) \tag{9}$$

where $\tilde{g}^u$ is the gradient after processing, the clipping function $\text{clip}(x, \sigma)$ constrains the gradient g^u within a range defined by σ. It ensures that the added noise does not excessively distort the original gradient. $\text{Laplacian}\,(0, \lambda)$ represents the Laplacian noise added to the clipped gradient. Laplacian noise is drawn from a Laplace distribution with mean 0 and scale parameter λ.

E. The Server-Side Multicenter Aggregation

In our federated recommendation framework, the server primarily performs three tasks: user selection, user clustering, and parameter aggregation. First, for the tth round, the server randomly selects a certain proportion of clients ρ to participate in the training. In a federated setting, the clients upload encrypted user–item embedding gradients and model gradients to the server. Then, the server conducts user clustering and parameter aggregation.

1) User Clustering: As described in the previous section, to address the issue of non-IID data and reduce communication costs, we intend to combine clustering models with global models to obtain personalized recommendation models. Before obtaining the clustering models, we first use the K-means algorithm to cluster users and assign similar users to the same group based on their embeddings. Specifically, we divide the users into K groups and assign each user n to its corresponding cluster $C(n)$. Each cluster represents a group of users with similar interests and behavior patterns.

2) Parameter Aggregation: Our framework presents two methods for aggregation, namely embedding aggregation and network model aggregation.

a) Embedding aggregation: User embeddings are stored locally and can be clustered through a trusted server after encryption. The item embeddings are jointly trained by all clients, and the item embedding gradients $\tilde{g}^u_i$ of user u are uploaded to the server for aggregation. We use the FedAvg for aggregation as follows:

$$\bar{g}_i = \sum_{u \in \mathcal{N}_t} \frac{m_u}{m} \tilde{g}^u_i \quad m = \sum_{u \in \mathcal{N}_t} m_u \tag{10}$$

where $\mathcal{N}_t$ is the set of clients selected in round t, m_u is the quantity of local datasets for client u, and $\bar{g}_i$ is the weighted average of item embedding gradients from the client side. After aggregation, gradient descent is used to update the item embeddings e^t_i as follows:

$$e^t_i = e^{t-1}_i - \eta \cdot \bar{g}_i \tag{11}$$

where η is the learning rate.

b) Network model aggregation: As for network models, we perform intragroup aggregation and intergroup aggregation to obtain the clustering model $h^t_{C(u)}$ and global model h^t_{global}. First, each group applies FedAvg to aggregate model parameters, resulting in the clustering model $h^t_{C(u)}$

$$\begin{aligned} \bar{g}_{C(u)} &= \sum_{u \in C(u)} \frac{m_u}{m} \tilde{g}^u_{C(u)} \quad m = \sum_{u \in C(u)} m_u \\ h^t_{C(u)} &= h^{t-1}_{C(u)} - \eta \cdot \bar{g}_{C(u)} \end{aligned} \tag{12}$$

where $C(u)$ is the cluster containing user u, and η is the learning rate. Since each cluster represents a set of similar clusters, to

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Algorithm 1: FedPCL

Require: Embedding Size d; Learning rate η; Clients loal graph $\{\mathcal{G}_n|_{n=1}^N\}$; Initialize $U = [e_n|_{n=1}^N], V = [e_i|_{i=1}^M], h_{global}$ at the server;

Ensure: Local client embeddings $U = [e_n|_{n=1}^N]$; Model parameters h;

1: // server
2: **while** not converge **do**
3: Cluster all users $\mathcal{U}$ into K clusters ▷ user clustering
4: Randomly select a subset $\mathcal{N}$ from the user set $\mathcal{U}$
5: **for** each $n \in \mathcal{N}$ **do**
6: $\tilde{g}_i^n, \tilde{g}_{C(n)} \leftarrow$ *ClientUpdate*(n, e_i^t , h_C , h_{global})
7: **end for**
8: $\bar{g}_i \leftarrow$ Eq. (10) ▷ your comment content
9: $e_i^t \leftarrow$ Eq. (11) ▷ item embeddings
10: $h_C^t \leftarrow$ Eq. (12) ▷ clustering model
11: $h_{global}^t \leftarrow$ Eq. (13) ▷ global model
12: **end while**
13: //client
14: **function** CLIENTUPDATE(n , e_i^t, h_C , h_{global})
15: Calculate the embedding gradients ∇U_n and g_i^n by Eq. (7)
16: Calculate the model gradients g_m^n by Eq. (8)
17: $e_n \leftarrow e_n - \eta \nabla U_n$ ▷ user embedding
18: $\tilde{g}_i^n, \tilde{g}_{C(n)} \leftarrow$ Eq. (9) ▷ LDP
19: **return** $\tilde{g}_i^n, \tilde{g}_{C(n)}$
20: **end function**

treat each cluster equally, we design a simple averaging method to aggregate the clustering models and obtain the global model

$$h_{\text{global}}^t = \sum_{n=1}^{K} \frac{1}{K} h_{C(n)}^t \tag{13}$$

where K represents the number of clusters. Then, the global model h_{global}^t and clustering model $h_{C(u)}^t$ are sent to user n for personalized recommendations.

F. FedPCL Algorithm

Algorithm 1 presents the pseudocode for the FedPCL algorithm. The input of this algorithm includes training hyperparameters such as embedding size d and learning rate η, as well as the personalized local graph network $\mathcal{G}_n$ for each client $n \in \{1, \ldots, N\}$, where N represents the number of clients. Additionally, initializing user embeddings $U = [e_n|_{n=1}^N]$, item embeddings $V = [e_i|_{i=1}^M]$, and model parameters h. Our objective is to provide personalized recommendations for each user, and the output consists of model parameters h and client-local embeddings e, which are used to predict ratings. In the algorithm, lines 2–12 represent a loop executed on the server. First, the server performs K-means clustering, grouping all users $\mathcal{U}$ into K clusters. After that, calculate the updated gradients $\bar{g}_i$ and e_i^t based on (10) and (11) and obtain clustering models h_C^t and global models h_{global}^t to provide personalized models for users. Then, parameters are sent to a randomly selected set of $\mathcal{N}$ clients, and their encrypted gradients are collected to update the parameters. The function ClientUpdate() represents the operation executed on the client side, where the client n computes BPR loss and structural contrastive loss using its local graph $\mathcal{G}_n$, and then the proposed framework updates the parameters accordingly. User embeddings e_n are stored locally, and item embedding gradients $\tilde{g}_i^n$ and model parameters $\tilde{g}_{C(n)}$ are uploaded to the server for updating after LDP encryption. This process is repeated until convergence.

IV. Experiments

A. Experimental Setup

In this section, we demonstrate the effectiveness of our method on five public datasets and compare our approach with existing baselines.

1) Datasets: In this article, we employ five publicly available real-world datasets, i.e., Filmtrust, Amazon-Electronic, Steam-200K (Steam), MovieLens-100K (ML-100k), and MovieLens-1M (ML-1M).

a) Filmtrust dataset: The Filmtrust dataset is a well-known public dataset that contains user ratings for movies. It comprises data from 1110 users and 1775 movies, with a total of 20 453 interactions.

b) Amazon-Electronic dataset: The Amazon-Electronic dataset consists of user ratings and reviews for electronic products available on Amazon. It comprises 1435 users, 1522 electronic products, and 35 931 interaction records.

c) Steam-200K dataset: The Steam-200K dataset is collected from the Steam gaming platform and contains 3753 users, 5134 games, and 114 713 interaction records.

d) MovieLens dataset: MovieLens-1M and MovieLens-100k are two popular movie recommendation datasets widely used in the field of recommender systems and machine learning. The MovieLens-100k dataset consists of 100 000 movie ratings collected from 943 users on 1682 movies. On the other hand, the MovieLens-1M dataset is a larger version, containing 1 000 209 ratings from 6040 users on 3706 movies.

During the preprocessing stage of our experiment, we convert various types of interactions into implicit feedback. To achieve a fair comparison, we use the leave-one-out method for evaluation. For each user, the last interaction is taken as the test set, and the items before the last interaction are used for training.

2) Baseline: To demonstrate the effectiveness of our method, we compared it with several competitive baseline methods.

a) FedAvg [10]: It is an algorithm based on federated learning that updates the global model by averaging the model parameters uploaded by participating clients.

b) FedMF [14]: It is a recommendation framework based on federated matrix factorization, which utilizes homomorphic encryption techniques to protect the gradients uploaded by users.

c) FedGNN [38]: It proposes a privacy-preserving federated recommendation framework based on GNN. It trains a global GNN model from distributed users and applies privacy protection techniques to prevent user information leakage.

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TABLE I
OVERALL PERFORMANCE COMPARISON OF THE PROPOSED FEDPCL AGAINST SIX BASELINES ON FIVE DATASETS

Dataset		Central	FedAvg	FedMF	FedGNN	PerFedRec	FPFR	FedPCL	Improv
ML-1M	HR@10	64.53	44.70	56.30	58.37	61.31	61.84	**62.86**	1.65%
	NDCG@10	46.78	24.90	31.30	39.84	42.83	43.62	**44.12**	1.15%
ML-100k	HR@10	64.31	42.70	58.34	60.30	61.87	62.59	**63.81**	1.95%
	NDCG@10	45.57	23.87	32.82	43.03	43.51	44.65	**45.03**	0.85%
Steam	HR@10	80.16	71.21	73.82	75.33	76.61	78.43	**80.36**	2.46%
	NDCG@10	66.03	50.22	54.09	62.09	62.63	64.76	**65.55**	1.22%
FilmTrust	HR@10	21.85	10.81	11.85	15.58	15.12	15.66	**16.81**	7.34%
	NDCG@10	10.53	4.83	5.82	8.17	8.01	8.04	**8.61**	5.39%
Amazon-Electronic	HR@10	38.83	26.53	28.22	30.40	32.64	32.76	**34.04**	3.91%
	NDCG@10	26.74	14.53	18.55	18.97	21.39	21.98	**22.93**	4.32%

Note: The best and second-best federated learning results are indicated by bold and underlined formatting, respectively. Results are shown in percentages (%), with evaluation metrics based on the top ten predictions from the test set.

d) Personalized Federated Recommendation (PerFedRec) [16]: It is an architecture based on GNN that designs personalized models for users based on their local data and achieves personalized recommendations in a federated setting.

e) FPFR [39]: It is an architecture based on federated GNN. It simultaneously considers both fairness and personalization in recommendation.

We also compare our model with a centralized learning approach LightGCN [8], which is referred to as center. To ensure fairness in the comparison, LightGCN is adopted as the backbone model of our proposed method and the baselines involving GNN architectures. Furthermore, all baselines are trained using the BPR loss function [41].

3) Evaluation Metrics: To evaluate and compare the performance of our proposed method, we adopt two metrics: hit rate at rank 10 (HR@10) and normalized discounted cumulative gain at rank 10 (NDCG@10). These metrics are widely used in the field of recommender systems to measure the accuracy and ranking quality of recommendation algorithms. The higher the scores are, the better the recommendation results are. In the experiments, we employ negative sampling for evaluation. For each user, we randomly sample 100 items that they have not interacted with. Subsequently, we combine the test items with the sampled negative samples and rank them.

4) Implementation Details: We use the Adam optimizer [45] for model training. In the proposed algorithm, the learning rate η is set to 0.1, the L2 regularization factor is set to $1e^{-6}$ and the learning loss weight λ is set to 1. The temperature parameter τ for contrastive learning is set to 0.3. Both hyperparameters μ_1 and μ_2 are set to 0.5, and the number of clusters K is set to 5. The communication rounds T are set to 400. For all methods, the embedding dimensions for users and items are both set to 64. The rating predictor adopts the dot product method. Each round selects 128 users to participate in model training. The contrastive learning loss will impact the convergence speed of the model. To enhance the model's convergence speed, we refrain from incorporating contrastive learning in the initial 20 epochs, allowing the model to converge rapidly during the early stages of training. To ensure the accuracy of experimental results, we conduct five rounds of training for all methods and provide average results.

B. Overall Performance

In this section, we demonstrate the recommendation performance of our proposed FedPCL method by comparing it with all the baselines. Table I presents the experimental results of FedPCL and all the baselines on five datasets. The best and the second-best results are bolded and underlined, respectively. Compared to the baselines in the federated setting, our proposed FedPCL outperforms the second-best results in terms of HR@10 and NDCG@10 on all five datasets. In all five datasets, the average improvement in HR@10 is 3.46%, while the average improvement in NDCG@10 is 2.59%.

Based on the results, we draw several key analyses. 1) Among all the methods in the federated setting, methods based on GNN (e.g., FedGNN, PerFedRec, FPFR, and FedPCL) outperform traditional federated recommendation methods (e.g., FedAvg and FedMF). This is because GNN can effectively leverage the graph structure information between users and items for recommendations. It propagates and aggregates information from neighboring nodes in the graph, allowing for better learning of high-order relationships in the user–item bipartite graph. 2) In the federated setting, our FedPCL method outperforms other federated methods across all datasets. This is attributed to our method considering the influence of data non-IID and designing personalized models for users to recommend. In contrast, FedMF, FedAvg, and FedGNN do not account for user personalization, employing the same global model for all users. They overlook the impact of non-IID data, leading to a decrease in model performance. Compared to PerFedRec and FPFR, which also consider model personalization, our method still outperforms them, which is attributed to our consideration of data sparsity in recommendation scenarios, utilizing structural contrastive learning to alleviate the impact of data sparsity and further enhance model performance. It demonstrates the effectiveness of our proposed personalized recommendation module and structural contrastive learning in improving recommendation performance. 3) Centralized methods almost

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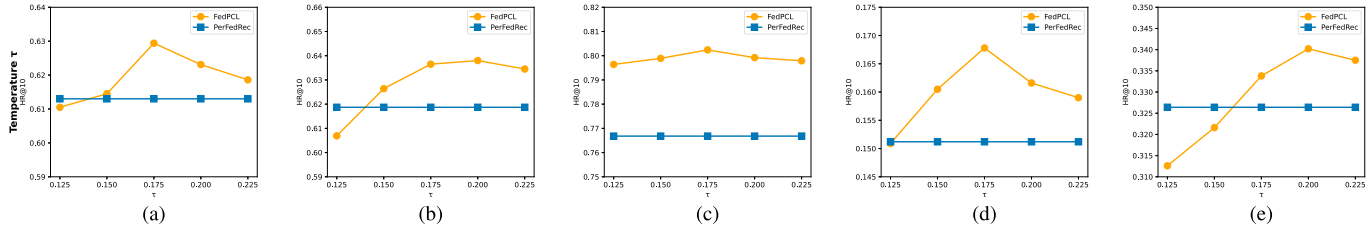

Fig. 3. Performance comparison of the proposed FedPCL and the baselines under different parameters. Top: Temperature τ. (a) ML-1M. (b) ML-100k. (c) Steam. (d) FilmTrust. (e) Amazon-Electronic.

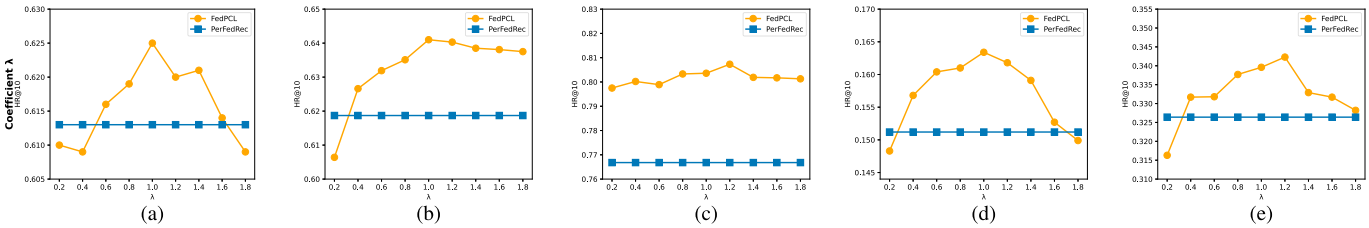

Fig. 4. Performance comparison of the proposed FedPCL and the baselines under different parameters. Middle: Coefficient λ. (a) ML-1M. (b) ML-100k. (c) Steam. (d) FilmTrust. (e) Amazon-Electronic.

always achieve the best results across all scenarios. There are two reasons for this. First, in a federated scenario, to protect user privacy, data are kept locally and the local data are sparse. Additionally, user data are non-IID, which limits the model's ability to capture global data patterns. Our FedPCL incorporates personalized recommendation and structural contrastive learning, which partially addresses this issue. Second, many privacy-preserving methods are employed in federated settings, such as FedPCL, FedGNN, and PerFedRec, which add random noise to protect uploaded local gradients. FedGNN designs a pseudointeraction sampling method, adding fake positive samples to the dataset to protect users' real interaction data. These privacy-preserving methods somewhat compromise the model's performance.

C. Parameter Sensitivity

1) Impact of the Temperature τ: To study the influence of the temperature τ defined in (5) and (6) on recommendation performance, we vary τ in the range of 0.125–0.225 and test performance on five datasets. The results are shown in Fig. 3. 1) The setting of the temperature coefficient τ in the contrastive loss function is crucial as it directly affects the model's performance. All five datasets perform best around $\tau \in \{0.175, 0.2\}$. 2) When τ is too small, recommendation performance significantly declines. The proposed ML-1M, ML-100K, and Amazon-Electronic datasets even show lower recommendation performance than the baseline at $\tau = 0.125$. Based on [46], this is because, with a smaller temperature coefficient τ, the contrastive learning effect becomes stronger, meaning similar sample distances become closer while dissimilar sample distances become farther. However, the model may mistakenly consider nodes closer to positive samples as negative samples, thereby increasing their distance. These negative samples might actually be potential positive samples. Hence, too small τ leads to a significant drop in model performance and generalization ability. 3) When τ is too large, recommendation performance of the proposed framework also decreases to some extent. This is because when τ is too large, the model struggles to distinguish between positive and negative samples, making it difficult to learn useful representations. In conclusion, choosing an appropriate temperature τ can make the distribution of nodes in the embedding space more uniform, achieving good recommendation results.

2) Impact of the λ: λ is defined in the structural contrastive loss (7). λ can adjust the influence ratio of user and item losses in the structural contrastive loss. We vary λ in the range of 0.2–1.8, and the experimental results are shown in Fig. 4. We can observe that: 1) proper λ can help the model to better learn user and item embeddings, effectively improving recommendation performance. When λ is around 1, the performance on all five datasets reaches the optimal. That is, when the ratio of contrastive losses between the user side and the item side is close, the performance is the best. This indicates that in our proposed FedPCL method, high-order information from both users and items is valuable. 2) For the Steam datasets, the performance differences are small across different λ values, but the recommendation performance is better than the baseline. This may be because their GNNs have fewer interconnected nodes and sparser interaction data. Structural contrastive learning can better learn node representations in sparse data, thus outperforming the baseline. The small performance differences indicate that this dataset is robust to the choice of λ. 3) When reducing the value of λ, the performance of FedPCL on the ML-1M, ML-100K, FilmTrust, and Amazon-Electronic datasets significantly declines, and even around λ of 0.2, the recommendation performance is lower than the baseline. This indicates that for these four datasets, the structural neighbors on the item side contain

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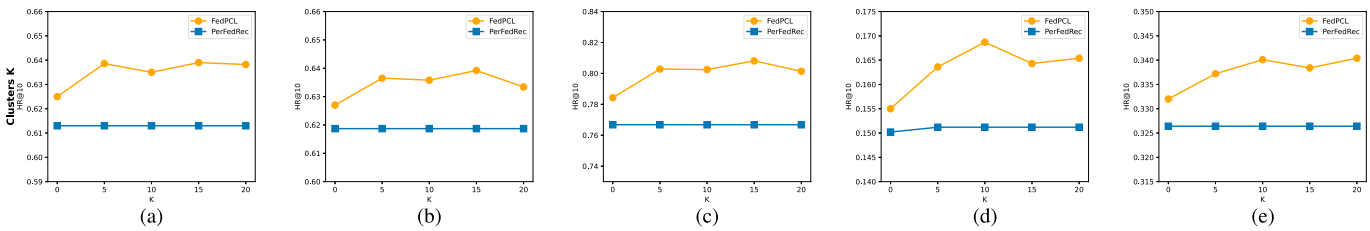

Fig. 5. Performance comparison of the proposed FedPCL and the baselines under different parameters. Bottom: Number of clusters K. (a) ML-1M. (b) ML-100k. (c) Steam. (d) FilmTrust. (e) Amazon-Electronic.

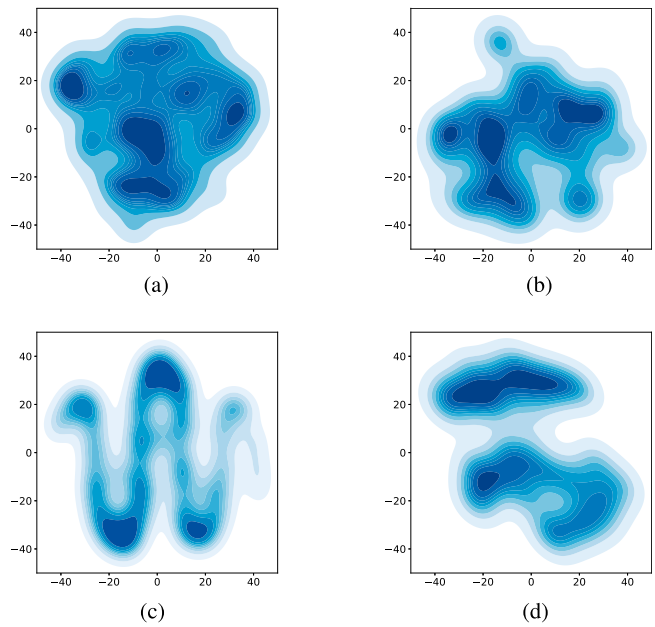

Fig. 6. Visualization of the feature distribution of item embeddings. The darker the color, the more items in the area. (a) FedPCL in ML-1M. (b) FedPCL in ML-100k. (c) FedGNN in ML-1M. (d) FedGNN in ML-100k.

rich information that can help the model to learn more accurate embeddings.

3) Number of Clusters K: To investigate the sensitivity of FedPCL to the number of clusters, we conduct experiments within the range of $K \in \{5, 10, 15, 20\}$ and set K to 0 to remove clustering. The experimental results are shown in Fig. 5. Clearly, regardless of the value of the cluster number K, FedPCL consistently outperforms the baseline. When we set the cluster number to 0, the performance on all five datasets significantly declines, indicating the effectiveness of our personalized recommendation module in enhancing recommendation performance. Additionally, it can be observed that the model's recommendation performance remains consistent across different cluster numbers, suggesting that the value of K is not a key factor in improving recommendation performance, thereby reducing the burden of adjusting hyperparameters.

D. Visualizing the Distribution of Representations

To better understand the effectiveness of our proposed method, we visualize the learned item embeddings on two datasets and compared them with FedGNN, as shown in Fig. 6.

TABLE II
ABLATION STUDY ON FIVE DATASET

Dataset		Variation 1		Variation 2		FedPCL
		Value	Diff	Value	Diff	Value
ML-1M	HR@10	59.18	3.58	60.44	2.32	62.76
	NDCG@10	41.87	2.37	42.95	1.29	44.24
ML-100k	HR@10	60.01	3.91	61.14	2.78	63.92
	NDCG@10	43.31	2.07	43.84	1.54	45.38
Steam	HR@10	76.92	3.35	78.43	1.84	80.27
	NDCG@10	62.71	2.61	63.53	1.73	65.32
FilmTrust	HR@10	15.36	1.47	16.02	0.81	16.83
	NDCG@10	8.17	0.75	8.57	0.35	8.92
Amazon-Electronic	HR@10	32.21	2.06	33.06	1.21	34.27
	NDCG@10	21.25	1.60	21.84	1.01	22.85

Note: The results are reported in percentage (%).

We utilize Gaussian kernel density estimation (KDE) to plot the learned embeddings in a 2-D space. From the density distribution perspective, the embeddings learned by FedGNN exhibit fewer high-density areas and a higher degree of overlap within these high-density regions. In contrast, FedPCL's embeddings are concentrated in the central region, forming a relatively centralized distribution with more high-density areas compared to FedGNN. Looking at the data distribution, FedGNN has fewer samples in the central region and a relatively scattered distribution. On the other hand, FedPCL's learned embeddings are more evenly distributed across the entire embedding space. This phenomenon indicates that FedGNN captures more important and extensive structures within the data, while our method better captures higher order relationships, effectively avoiding excessive concentration or overlap of embeddings, resulting in a more uniform distribution with highlighted importance. This allows the learned embeddings to better reflect the features of different nodes. We speculate that a more uniform distribution facilitates more accurate differentiation of interest differences among users or feature differences among items, leading to more effective recommendations. This is attributed to our structured-contrastive learning method, which better captures the structural neighbor information of nodes, thereby enhancing the quality of node embeddings.

E. Ablation Study

We conduct an ablation study on five datasets to validate the effectiveness of each module and assess their contributions

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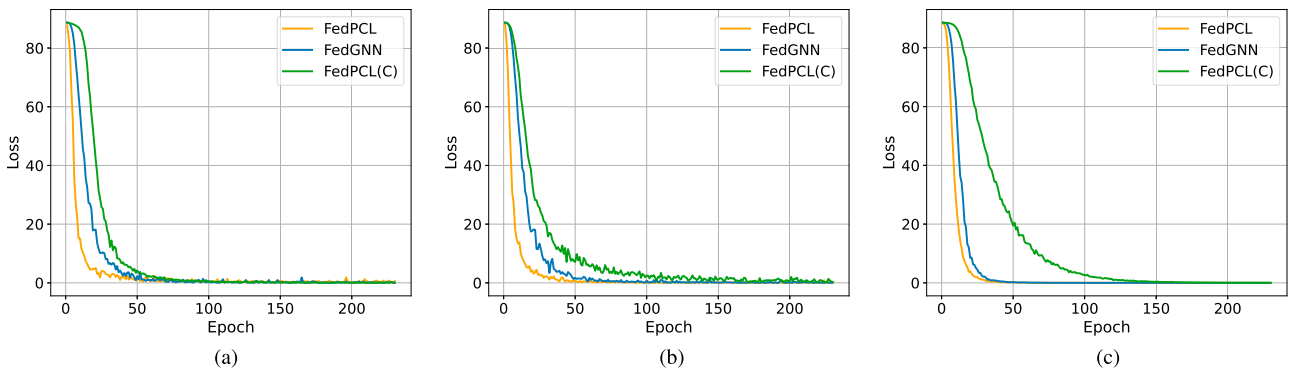

Fig. 7. Loss curve w.r.t training epoch on three datasets: (a) ML1M; (b) Steam; and (c) Amazon-Electronic.

to performance. Specifically, we create two variants of the FedPCL model. Variation1 removes the contrastive learning module, utilizing BPR loss for model training while retaining the personalized recommendation part. Variation2 removes the personalized recommendation module, and all clients utilize the same recommendation model, optimizing the model using BPR and structural contrastive loss.

The performance comparison of these two variants on five datasets is shown in Table II, along with the results of FedPCL and the performance differences between it and the variants. 1) Both variants outperform the baseline FedGNN. Additionally, our FedPCL model demonstrates better performance than variants with single module. This indicates that the proposed personalized recommendation module and structural contrastive learning module contribute to enhancing the model's recommendation performance, and they complement each other in enhancing performance. This is because the structural contrastive learning module can better learn node representations, improving the accuracy of node clustering. The personalized module is based on clustering, thereby enhancing the performance of cluster-level models in the personalized module. The purpose of the personalized module is to alleviate the impact of non-IID data to some extent, thereby further improving the model's recommendation performance. 2) Further analysis reveals that the difference between Variation2 and FedPCL is smaller compared to Variation1. In other words, the most significant performance improvement in the five datasets comes from the structural contrastive learning module. This is because, as discussed in Section III, in the context of recommendation systems, structural contrastive learning is not affected by data sparsity. It reduces the distance between adjacent nodes in the graph network and helps the model to better capture higher order information of neighboring nodes, thereby learning more accurate node representations and improving recommendation performance.

F. Convergence Study

To investigate the convergence of our proposed algorithm and explore the impact of contrastive learning and personalized modules on convergence, we plot the convergence curves of FedPCL, FedGNN, and FedPCL(C) on three datasets in Fig. 7. Here, FedPCL(C) refers to the variant of FedPCL with the personalized module removed and only applies contrastive learning. It can be observed that on the ML-1M dataset, FedPCL, FedGNN, and FedPCL(C) converge around epochs 80, 100, and 110, respectively. On the Steam dataset, FedPCL, FedGNN, and FedPCL(C) converge around epochs 70, 100, and 180, respectively. On the Amazon-Electronics dataset, FedPCL, FedGNN, and FedPCL(C) converge around epochs 45, 55, and 160, respectively. Through analysis, compared to FedGNN, FedPCL(C) demonstrates slower convergence on all three datasets. This is due to the principle of contrastive learning is to shorten the distance between similar nodes. During the early stages of model training, the learned node representations are not yet accurate. At this point, contrastive learning reduces the optimization speed, requiring more time to explore the parameter space and find the optimal solution. To address this, we introduce the personalized module. Compared to FedGNN, FedPCL shows faster convergence on all three datasets. This indicates that, compared to traditional federated recommendation algorithms, our proposed method has higher training efficiency, significantly reducing the training epochs. This reduction in epochs lowers communication overhead in federated recommendation scenarios. This improvement is attributed to the introduction of the personalized module. Within the personalized module, we design a clustering model to cluster privacy-protected users and jointly train the clustering model. This enables user b to directly learn the model parameters of user a within the same cluster, accelerating the training process for user b and mitigating the impact of contrastive learning on training speed. Therefore, the overall training efficiency of the proposed federated recommendation algorithm improves.

V. Conclusion

In this work, we proposed a new federated learning recommendation framework. Unlike other federated recommendation algorithms, we simultaneously addressed the issues of data sparsity and personalization within federated learning. We present the structural contrastive learning, which looks at the neighbors of nodes from a graph structure perspective. We also leveraged structural neighbors on GNNs to construct contrastive pairs, pulling neighboring nodes closer in the

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embedding space. This approach better captures the potential relationships between nodes, enhancing recommendation performance. To tackle the challenge of heterogeneity, we clustered similar users and combined global models with clustered models to achieve personalized recommendations. Additionally, clustered model is introduced to improve the model's convergence rate and reduced communication overhead. Experimental results on five real datasets proved the effectiveness of our proposed FedPCL.

In this article, we employed a federated learning architecture, where user data do not need to be centrally stored and only the parameters of the user models are uploaded. This aggregation method protects user privacy and holds significant importance for the application of recommendation systems in the field of privacy protection. However, our proposed model still has some weaknesses. One apparent challenge is how to further extend our method to more complex datasets and multimodal data to improve the performance of personalized recommendations. Currently, FedPCL primarily relies on single-modal data, such as user–item interaction data, which may be insufficient when dealing with complex high-dimensional datasets or multimodal data. Therefore, future research will focus on integrating more types of data, such as textual descriptions, image features, and audio information, to capture the comprehensive characteristics of users and items. Additionally, our research is based on static user–item interaction data. However, user preferences are often not static but change over time in the real world. Traditional static models may struggle to adapt to these changes, leading to a gradual decline in recommendation effectiveness. To address this issue, future research can explore dynamic learning and adaptive mechanisms that enable models to continuously update their understanding of user preferences in real time.

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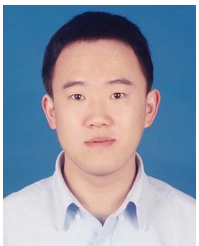

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